

Expected Output

Task 1

```
(.venv) ubuntu@ip-10-192-1-51:~$ nano de300_hw2.py
(.venv) ubuntu@ip-10-192-1-51:~$ python3 de300_hw2.py --tasknum 1
File does not exist in bucket. Will download.
Uploaded
-
```

Task 2

```
(.venv) ubuntu@ip-10-192-1-51:~$ python3 de300_hw2.py --tasknum 2
embeddings do not exist in bucket. Will create and upload.
Downloaded zip file from s3 bucket
collecting users from ratings.dat
users total=6040, sampled=1812
collecting movie_ids for sampled users
reading movies.dat
finished movies.dat
Number of movies embedded: 3529
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN
Loading weights: 100%|██████████| 100/100 [00:00<00:00, 1357.57it/s, Materializing par
DistilBertModel LOAD REPORT from: distilbert-base-uncased
Key | Status |
-----+-----+
vocab_transform.bias | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_layer_norm.bias | UNEXPECTED |
vocab_transform.weight | UNEXPECTED |
vocab_projector.bias | UNEXPECTED |
```

Notes:

```
- UNEXPECTED :can be ignored when loading from different task/architecture; not
embedding batch starting at 0/3529
[embedding batch starting at 200/3529
embedding batch starting at 400/3529
embedding batch starting at 600/3529
embedding batch starting at 800/3529
embedding batch starting at 1000/3529
embedding batch starting at 1200/3529
embedding batch starting at 1400/3529
embedding batch starting at 1600/3529
embedding batch starting at 1800/3529
embedding batch starting at 2000/3529
embedding batch starting at 2200/3529
embedding batch starting at 2400/3529
embedding batch starting at 2600/3529
embedding batch starting at 2800/3529
embedding batch starting at 3000/3529
embedding batch starting at 3200/3529
embedding batch starting at 3400/3529
movie subset embeddings shape: torch.Size([3529, 768])
```

Task 3

Task 4

```
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
embedding batch starting at 0/3706
embedding batch starting at 200/3706
embedding batch starting at 400/3706
embedding batch starting at 600/3706
embedding batch starting at 800/3706
embedding batch starting at 1000/3706
embedding batch starting at 1200/3706
embedding batch starting at 1400/3706
embedding batch starting at 1600/3706
embedding batch starting at 1800/3706
embedding batch starting at 2000/3706
embedding batch starting at 2200/3706
embedding batch starting at 2400/3706
embedding batch starting at 2600/3706
embedding batch starting at 2800/3706
embedding batch starting at 3000/3706
embedding batch starting at 3200/3706
embedding batch starting at 3400/3706
embedding batch starting at 3600/3706
movie subset embeddings shape: torch.Size([3706, 768])
Uploaded recommendations to s3://fingerson-winter26hw2/T4_full_recommendations.csv
  User_ID  User_Type Last_Interaction_Time  ...  Recommended_Movies  Recommended_Movie_Titles Top
0         cold_user  ...  2858 | 260 | 1196 | 1210 | 480  American Beauty (1999) | Star Wars: Episode IV...
1    4682    top_user      999055052    ...  3727 | 2551 | 3495 | 2148 | 431  Near Dark (1987) | Dead Ringers (1988) | Roads...

[2 rows x 7 columns]
```

Task 5

```
((.venv) ubuntu@ip-10-192-1-51:~$ python3 de300_hw2.py --tasknum 5
profile does not exist in bucket. creating and uploading.
Uploaded s3://fingerson-winter26/hw2/livs_profile.csv
  movie_id  rating
0        296      5
1        2571     4
2        2021     3
3        3034     4
4        2495     4
5        2810     3
6        3897     5
7        1582     4
8         592     3
9        3702     4
uploaded personalized recs to s3://fingerson-winter26/hw2/hw2/T5_personalized_recommendations.csv
  User_ID  Profile_Key  ...  Recommended_Movies  Recommended_Movie_Titles
0  my_profile  hw2/livs_profile.csv  ...  198 | 3802 | 2407 | 2174 | 2808  Strange Days (1995) | Freejack (1992) | Cocoon...
```

[1 rows x 6 columns]